NATIONAL ECONOMICS UNIVERSITY

Comparative Analysis of ETS and SARIMAX Models for Forecasting U.S. International Air Traffic

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1 Introduction

Forecasting international air traffic is an important task for governments, airlines, and airport authorities because accurate predictions help support transportation planning, resource allocation, and economic decision-making. Air passenger demand usually exhibits strong seasonal patterns, long-term growth trends, and unexpected disruptions caused by external events such as economic crises or the COVID-19 pandemic. Therefore, selecting an appropriate time series forecasting model is essential for improving prediction accuracy and understanding the behavior of air traffic data.

This study focuses on analyzing and forecasting U.S. international air traffic using two publicly available datasets: International_Report_Passengers.csv and International_Report_Departures.csv. The datasets contain monthly records of international passenger volumes and flight departures between U.S. airports and foreign destinations from 1990 to 2020. By aggregating the data into monthly time series, this research investigates the temporal patterns, trends, and seasonal characteristics present in international aviation activity.

The main objective of this paper is to compare the forecasting performance of two classical time series approaches: the Exponential Smoothing State Space (ETS) model and the Seasonal AutoRegressive Integrated Moving Average with eXogenous variables (SARIMAX) model. ETS is widely used for modeling trend and seasonality in univariate time series, while SARIMAX extends the SARIMA framework by incorporating seasonal autoregressive behavior and external explanatory variables. In this study, both models are trained and evaluated using historical air traffic data, and their forecasting performance is assessed through evaluation metrics including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2).

The contribution of this research is twofold. First, it provides a comparative analysis of ETS and SARIMAX models on real-world aviation datasets containing strong seasonal fluctuations and structural shocks. Second, it identifies the strengths and limitations of each model in forecasting international air traffic demand, particularly during periods of abnormal changes such as the COVID-19 outbreak. The findings of this study may support future research and practical forecasting applications in the aviation and transportation sectors.

2 Data and Methodology

At first, I try to test some time-series aspects on the data. They include : seasonal decomposition, heteroscedasticity, stationary , shock/break detection.
The stationarity analysis for both the Passengers and Departures datasets indicates that the time series are non-stationary. For the Passengers series, the Augmented Dickey–Fuller (ADF) test produced a p-value of approximately 0.327, while the KPSS test returned a p-value below 0.01. Similarly, the Departures series showed an ADF p-value of approximately 0.274 and a KPSS p-value below 0.01. Since the ADF test failed to reject the null hypothesis of a unit root and the KPSS test rejected the null hypothesis of stationarity in both cases, the results consistently confirm the presence of non-stationarity.Therefore, additional preprocessing steps such as logarithmic or Box–Cox transformation and differencing are required before fitting forecasting models like SARIMAX or Exponential Smoothing.
The heteroscedasticity is checked and the result is not a suprise.Since the p-value perspectively

0.000000 and 0.000033 for Passengers and Departures, both two data is heteroscedasticity, which means that P < 0.05 and we reject the null hypothesis. So we need to log transformation for target before training.
Shock/Break Detection "caught" three timeline : the first is 2020 (when the virus was spreaded all over the world) , the financial crisis in 2008 and the terrorist attack in 2001. When i check the outlier , i recognized the function can remove these timeline since ít value go beyond the threshold(z_thres >3), so i decide to created dummy variable for these events to keep the structural shocks usable for SARIMAX.

3 Feature Engineering

1. Calendar and Temporal Dynamics

- **Temporal Features (`add_temporal_features`)**: The code extracts absolute time components (year, month, quarter) and boundary indicators (start/end of month or year). This helps the model capture localized calendar effects (e.g., end-of-year holiday rushes).
- **Cyclical Encoding (`add_cyclical_features`)**: This is a highly advanced and mathematically sound technique. By applying sine and cosine transformations to the `month` and `quarter` variables, it converts a linear scale (1 to 12) into a circular continuum. This allows the model to understand that December (12) and January (1) are chronologically adjacent, preventing artificial jumps in the data.

2. Autoregressive and Momentum Indicators

While SARIMA inherently handles lags, explicitly creating these as features can be highly beneficial when combining SARIMAX with machine learning models (like XGBoost) or providing additional baseline context:

- **Lag Features (`add_lag_features`)**: Extracts exact historical passenger volumes from 1 month, 1 quarter (3 months), and 1 year (12 months) prior.
- **Rolling Statistics (`add_rolling_features`)**: Computes moving averages and rolling standard deviations. The 3-month moving average captures short-term momentum, while the 12-month acts as a long-term trend baseline. The rolling standard deviation acts as a proxy for market volatility.
- **Growth Features (`add_growth_features`)**: Computes Month-over-Month (MoM) and Year-over-Year (YoY) percentage changes. This shifts the model's focus from absolute passenger numbers to the rate of acceleration or deceleration, which is critical for forecasting economic cycles.

3. Structural Shocks and Anomaly Handling

- **Shock Features (`add_shock_features`)**: This directly addresses the "Black Swan" events we discussed previously. By creating binary dummy variables (`covid_dummy` for 2020-2021 and `post_covid_recovery` for 2022 onwards), the pipeline explicitly tells the SARIMAX model when a structural break occurs. This prevents the model's core trend and seasonality parameters from being corrupted by the massive drops caused by the pandemic.

4. **Aviation Domain-Specific Features**

- **Composition Features (`add_composition_features`)**: Calculates the ratio of Scheduled versus Charter flights. This provides business context, as a sudden spike in charter ratios might indicate irregular demand (e.g., military transport, rescue flights, or sports events) rather than stable commercial growth.

- **Route Features (`add_route_features`)**: Concatenates origin, destination, and carrier codes. While not directly usable in a standard univariate SARIMAX, this is an excellent foundation if you plan to scale up to a Panel Data Forecasting approach (e.g., forecasting multiple specific routes simultaneously).

5. **Variance Stabilization**

- **Log Transformation (`add_log_feature`)**: Uses `np.log1p` (logarithm plus one to handle zeros). As identified in your earlier Breusch-Pagan test, aviation data often suffers from heteroscedasticity (variance increases as total volume increases over the years). The log transformation compresses this variance, forcing the data into the "homoscedastic" state required by OLS/ARIMA-based algorithms for stable predictions.

6. **Execution and Model Integration**

- **The Orchestrator (`sarimax_feature_engineering`)**: Wraps all the aforementioned functions into a clean, sequential pipeline with print statements for logging. This ensures reproducibility.

- **Feature Selection (`get_exogenous_features`)**: Acts as a crucial filter. It drops string variables (like route names or dates) and target variables, retaining only the pure numerical matrix needed for the `exog` parameter in the `statsmodels` library.

- **Implementation**: The final example correctly applies the pipeline, isolates the `exog` matrix, sets the `log_Passengers` as the target variable `y`, and feeds them into the SARIMAX algorithm.

4 Result

The empirical evaluation strongly validates the theoretical advantage of SARIMAX over the traditional ETS model when handling time series with structural breaks. SARIMAX significantly outperformed ETS, achieving a much higher accuracy with a Mean Absolute Percentage Error (MAPE) of just 7.15% compared to ETS's 12.08%. Furthermore, the MAE and RMSE metrics for SARIMAX (1.28M and 1.46M, respectively) were nearly half of those produced by the ETS model (2.24M and 2.75M).

This predictive superiority is directly attributed to SARIMAX's capability to integrate exogenous dummy variables, which successfully isolated historical anomalies. The model's coefficients reveal the sheer magnitude of these black swan events: the COVID-19 pandemic caused a devastating direct drop of approximately 14.64 million passengers (-14,642,398), while the 9/11 attacks and the 2008 financial crisis accounted for drops of 3.02 million and 549 thousand passengers, respectively. By absorbing these extreme shocks through dummy variables, SARIMAX successfully protected its core trend and seasonal components from distortion, whereas the ETS model—blind to exogenous factors—was severely dragged down by the historical data collapse, leading to less accurate forecasts."

5 Implications

The superiority of the SARIMAX model over ETS in handling structural shocks carries profound implications for strategic management and planning within the aviation sector. The findings of this study directly impact three critical operational and financial areas for airlines and airport authorities:

1. Optimizing Resource Allocation Avoiding Post-Crisis Shortfalls Aviation is a highly capital-intensive industry. If airlines rely on rigid forecasting systems (like ETS) to plan post-pandemic operations, they will be misled by the historical collapse in data, leading to severe under-forecasting of future demand. Consequently, they risk facing critical shortages of active aircraft, pilots, cabin crew, and ground staff when the market rapidly rebounds. By utilizing SARIMAX to "isolate" the pandemic variable, airlines can uncover the true underlying "baseline" of natural demand. This empowers them to confidently reactivate their fleets ahead of the curve and capture market share from slower-moving competitors.

2. Financial Risk Management Against "Black Swan" Events Successfully modeling macro-shocks such as 9/11, the 2008 Financial Crisis, and COVID-19 allows executives to accurately quantify the sheer financial and operational "cost" of a disaster. For example, isolating the fact that COVID-19 directly wiped out approximately 14.6 million passengers provides an invaluable quantitative foundation for Chief Financial Officers (CFOs). This shock-adjusted data is crucial for conducting rigorous stress-testing, calculating the necessary scale of emergency cash reserves, and adjusting fuel hedging strategies to ensure corporate survival should a similar global crisis strike again.

3. Long-Term Airport Infrastructure Planning For airport authorities, expanding terminals or constructing new runways requires a 10-to-20-year strategic vision. If historical time series data is not cleaned and structural shocks are not extracted using dummy variables, the long-term growth trend becomes heavily distorted downward. This distortion can lead to the dangerous postponement of vital infrastructure projects. SARIMAX successfully protects the core "Trend" component from these breaks, providing airport managers with the reliable, shock-free data required to justify and negotiate multi-billion-dollar infrastructure loans with governments and financial institutions.

6 Conclusion

Ultimately, in a macro-sensitive industry like aviation, selecting the appropriate time series forecasting model is far more than a mere statistical exercise. It acts as the strategic compass that dictates the success or failure of billion-dollar investments and determines the resilience of an entire transportation ecosystem.